

Pranjal Rawat

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EDUCATION

Georgia Institute of Technology

Atlanta, GA

Bachelor of Science in Computer Science

Expected Graduation: May 2028

- Relevant coursework: Data Structures and Algorithms, Object-Oriented Programming, Computer Organization, Linear Algebra, Discrete Mathematics for CS, Multivariable Calculus

TECHNICAL SKILLS

Languages: Python, SQL, JavaScript, HTML/CSS, Java

Frameworks & Tools: Spring Boot, ReactJS, Next.js, Node.js, Streamlit, AWS (EC2, S3), Git, VS Code

Libraries: Pandas, NumPy, SciPy, PyArrow, Matplotlib, Scikit-learn, PyTorch, OpenCV, LangChain, Hugging Face

AI/ML: RAG pipelines, vector databases (ChromaDB), multi-agent orchestration, YOLOv8, Vision Transformers

EXPERIENCE

Data Analyst Researcher

August 2026 – Present

Georgia Tech Women's Basketball

Atlanta, GA

- Collaborating with Georgia Tech's Division 1 Women's Basketball team to **generate data-driven insights using computer vision and ML models**, integrating Roboflow APIs and SDKs into scalable pipelines.
- Building YOLOv8 and Vision Transformer (ViT) models in Python, PyTorch, and OpenCV for object detection and action classification on game footage.
- Applying regression and clustering models to individual and team performance data to support scouting and in-game strategy decisions.

Data Science Intern

June 2026 – August 2026

ThinkDigits Inc.

Remote

- **Built a sensor analytics pipeline for a 47-machine PVC extrusion line** by processing 2.5M+ minute-level readings in Python, surfacing the top failure modes driving \$1M+ in downtime cost.
- Raised verified data coverage from 49% to 94% by applying rolling Mann-Whitney U and Kolmogorov-Smirnov changepoint tests on per-machine vibration data to detect product transitions, unlocking clean per-item sensor data for downstream OEE and predictive-maintenance modeling.
- Scaled a machine-health anomaly-detection framework from 12 to 47 production assets by implementing a robust Median/MAD baselining method across sensor-profile-based machine groups, enabling per-cluster thresholds for the predictive maintenance pipeline.

AI Engineer Intern

January 2026 – May 2026

TechX

- Architected a Startup Idea Validator Agent that breaks a business idea into four specialized analyses (market sizing, competitor intelligence, financial projections, and risk evaluation) coordinated through multi-agent orchestration into a structured report with a weighted viability score.
- **Engineered a RAG pipeline over 40+ U.S. market and competitor documents**, embedding them into a vector database (ChromaDB) so agents ground answers in retrieved sources rather than model recall, raising relevance on a 25-case eval set from 62% to 88%.
- Built a 25-case auto-graded test suite that uses an LLM to judge answers against expected results, with positive and negative cases to catch hallucinated or incorrect responses and validate quality after each change to the data or model.

PROJECTS

PantryUp | *Next.js, React, Node.js, JavaScript, TheMealDB API, Groq API (Llama 3.3), Vercel*

- Built a full-stack recipe search app unifying a public REST API and a 16,500-row CSV dataset behind a provider layer with a normalized schema, plus a Groq-powered natural-language search that lets users query by pantry ingredients instead of recipe names.

MednCare | *Python, ReactJS, Google Gemini API, Chart.js, JavaScript, CSS*

- Built a personal healthcare web app for managing prescriptions and health records, with a dashboard for trend visualization and doctor-ready reports, a medication timer with live countdowns and browser notifications, and a Gemini-powered AI assistant that answers dosage and schedule questions (labeled as non-professional guidance).